Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The remainder when $6^n - 5n$ is divided by 25, is

(A) 1

(B) 24

(C) 0

(D) n

Q.2

The remainder when 2^{2000} is divided by 17 is

(A) 1

(B) 2

(C) 8

(D) 12

Q.3

For each $n \in \mathbf{N}$, $2^{3n} - 1$ is divisible by

(A) 8

(B) 16

(C) 32

(D) none of these

Q.4

For all $n \in \mathbf{N}$, $n(n+1)(2n+1)$ is divisible by

(A) 6

(B) 8

(C) 15

(D) 9

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Q.5

If $a_n = 2^{2^n} + 1$, then for $n > 1$, last digit of a_n is

(A) 5

(B) 7

(C) 3

(D) 4

Q.6

Statement-1: For each natural number n, $(n+1)^7 - n^7 - 1$ is divisible by 7
Statement-2: For each natural number n, $n^7 - n$ is divisible by 7.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.7

For each $n \in \mathbf{N}$, $x^{2n-1} + y^{2n-1}$ is divisible by

(A) $x + y$

(B) $(x+y)^2$

(C) $x^3 + y^3$

(D) none of these

Q.8

If $x = 99^{50} + 100^{50}$ and $y = (101)^{50}$ then

(A) $x = y$

(B) $x < y$

(C) $x > y$

(D) None of these

Q.9

The greatest integer less than or equal to; $(\sqrt{2}+1)^6$ is

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(A) 196

(B) 197

(C) 198

(D) 199

Q.10

If $y = 3x + 6x^2 + 10x^3 + \ldots\ldots\ldots$, then
$\frac{1}{3}y - \frac{1.4}{3^2 2}y^2 + \frac{1.4.7}{3^2 3}y^3 - \ldots\ldots\infty$ is equal to

(A) x

(B) $1 - x$

(C) $1 + x$

(D) x^x

Q.11

If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{\frac{3}{2}} - \left(1+\frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{2}}}$ may be approximated as

(A) $1 - \frac{3}{8}x^2$

(B) $3x + \frac{3}{8}x^2$

(C) $-\frac{3}{8}x^2$

(D) $\frac{x}{2} - \frac{3}{8}x^2$

Q.12

The remainder when 27^{40} is divided by 12 is

(A) 3

(B) 7

(C) 9

(D) 11

Q.13

The minimum positive integral value of m such that $(1073)^{71} - \mathrm{m}$ may be divisible by 10 , is

(A) 1

(B) 3

(C) 7

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(D) 9

Q.14

For every natural number n, $3^{2n+2} - 8n - 9$ is divisible by

(A) 16

(B) 128

(C) 256

(D) none of these

Q.15

The positive integer just greater than $(1 + 0.0001)^{10000}$ is

(A) 4

(B) 5

(C) 2

(D) 3

Q.16

If $7^9 + 9^7$ is divided by 64 then the remainder is

(A) 0

(B) 1

(C) 2

(D) 63

Q.17

The fractional part of $\frac{2^{4n}}{15}$ is

(A) $\frac{1}{15}$

(B) $\frac{2}{15}$

(C) $\frac{4}{15}$

(D) None of these

Q.18

$\sqrt{5}\left[(\sqrt{5}+1)^{50} - (\sqrt{5}-1)^{50}\right]$ is

(A) anirrational number

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(B) 0

(C) a natural number

(D) None of these

Q.19

If $x = (2+\sqrt{3})^n$, then find the value of $x(1-\{x\})$, where $\{x\}$ denotes the fractional part of x

(A) 1

(B) 2

(C) 2^{2n}

(D) 2^n

Q.20

If 7^{103} is divided by 25, then the remainder is

(A) 20

(B) 16

(C) 18

(D) 15

Answers & Solutions

Q.1 Answer:

1

Solution:

$6^n - 5n = (1+5)^n - 5n$
$= 1 + 5n + {}^nC_2\left(5^2\right) + {}^nC_3\left(5^3\right) + \ldots + {}^nC_n\left(5^n\right)$
$-5n$
$= 1 + 25m$
where $m = {}^nC_2 + {}^nC_3(5) + \ldots + {}^nC_n\left(5^{n-2}\right)$
is a whole number. Thus, remainder when $6^n - 5n$ is divided by 25 is 1.

Q.2 Answer:

1

Solution:

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We have $2^4 = 16 = 17 - 1$

$$\begin{aligned}\therefore \quad 2^{2000} &= \left(2^4\right)^{500} = (17-1)^{500} \\ &= {}^{500}C_0 17^{500} - {}^{500}C_1 17^{499} + {}^{500}C_2 17^{498} - \\ &\quad {}^{500}C_3 17^{497} + \ldots - {}^{500}C_{499}(17) + (-1)^{500} \\ &= 17m + 1\end{aligned}$$

where m **is some positive integer. Thus, the required remainder is 1.**

Q.3 Answer:

none of these

Solution:

We have

$$\begin{aligned}2^{3n} - 1 &= \left(2^3\right)^n - 1 = (1+7)^n - 1 \\ &= 1 + {}^nC_1(7) + {}^nC_2\left(7^2\right) + \ldots + {}^nC_n\left(7^n\right) - 1 \\ &= 7m\end{aligned}$$

where $m = {}^nC_1 + {}^nC_2(7) + \ldots + {}^nC_n\left(7^{n-1}\right)$ **is a positive integer.**

Thus, 7 divides $2^{3n} - 1$ **for all** $n \in \mathbf{N}$**.**

Q.4 Answer:

6

Solution:

We have $n(n+1)(2n+1) = 6\left\{1^2 + 2^2 + \ldots + n^2\right\}$

$\therefore \quad n(n+1)(2n+1)$ **is divisible by 6 for all** $n \in \mathbf{N}$**.**

Q.5 Answer:

7

Solution:

For $n = 2, a_n = 2^4 + 1 = 16 + 1 = 17$**. Assume that** $a_k = 2^{2^k} + 1 = 10m + 7$ **where** $k > 1$ **and** m **is some positive integer.**

Now,

$$\begin{aligned}a_{k+1} &= 2^{2^{k+1}} + 1 = \left(2^{2^k}\right)^2 + 1 \\ &= (10m+6)^2 + 1 = 10\left(10m^2 + 12m + 3\right) + 7\end{aligned}$$

Thus, last digit of a_n **is** $7 \forall n > 1$**.**

Q.6 Answer:

STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

Solution:

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For Statement-2, write $n = 7r + k, k = -3, -2, -1, 0, 1, 2, 3$

Now, $n^7 - n = (7r + k)^7 - (7r + k)$

Using binomial theorem, we get $n^7 - n - \left(k^7 - k\right)$ is divisible by 7.

$\Leftrightarrow$ $n^7 - n$ is divisible by 7 if and only if $k^7 - k$ is divisible by 7 For $k = \pm 3, \pm 2, \pm 1, k^7 - k$ is divisible by 7 if and only if $k^6 - 1$ is divisible by 7.

But $k^6 - 1 = 728, 63, 0$ for $k = \pm 3, \pm 2, \pm 1$

$\therefore$ $k^6 - 1$ is divisible by 7 for $k = \pm 3, \pm 2, \pm 1$.

Thus, $n^7 - n$ is divisible by 7 for each $n \in \mathbf{N}$. $\Rightarrow$ $(n + 1)^7 - (n + 1) - \left(n^7 - n\right)$ is divisible by 7

$\Rightarrow$ $(n + 1)^7 - n^7 - 1$ is divisible by 7 . Therefore, both the statements are true and statement - 2 is a correct reason for Statement-1.

Q.7 Answer:

$x + y$

Solution:

$x^{2n-1} + y^{2n-1}$

$= (x + y)\left(x^{2n-2} - x^{2n-3}y + x^{2n-4}y^2 - x^{2n-5}y^3 + \ldots + y^{2n-2}\right)$

Thus $x^{2n-1} + y^{2n-1}$ is divisible by $x + y$ for all $n \in \mathbf{N}$.

Q.8 Answer:

$x < y$

Solution:

$$(101)^{50} - (99)^{50} = (100 + 1)^{50} - (100 - 1)^{50}$$
$$= 2\left[{}^{50}\mathrm{C}_1(100)^{49} + {}^{50}\mathrm{C}_3(100)^{47} + \ldots\ldots + {}^{50}\mathrm{C}_{49}(100)\right]$$
$$> 2 \cdot {}^{50}\mathrm{C}_1 \cdot (100)^{49} = 2 \times 50(100)^{49} = (100)^{50}$$
$$\Rightarrow (101)^{50} > (99)^{50} + (100)^{50} \Rightarrow \mathrm{y} > \mathrm{x} \Rightarrow \mathrm{x} < \mathrm{y}.$$

Q.9 Answer:

197

Solution:

$(\sqrt{2} + 1)^6 = \mathrm{A}$

$[\mathrm{A}]$ is the greatest integer less than

Let f be some unknown fraction.

Hence, we can write $[A] + f = A$.

Now, $(\sqrt{2} - 1) < 1$

$(\sqrt{2} - 1)^6 < 1$

Let, $(\sqrt{2} - 1)^6$ be denoted by f_1.

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So, $(\sqrt{2}+1)^6+(\sqrt{2}-1)^6=[\mathrm{A}]+\mathrm{f}+\mathrm{f}_1$

$\Rightarrow 2\left({}^6\mathrm{C}_0\cdot(\sqrt{2})^6+{}^6\mathrm{C}_2\cdot(\sqrt{2})^4+{}^6\mathrm{C}_6\right)=[\mathrm{A}]+\mathrm{f}+\mathrm{f}_1$

$\Rightarrow 2(8+60+31)=[\mathrm{A}]+\mathrm{f}+\mathrm{f}_1$

$\Rightarrow 198=[\mathrm{A}]+\mathrm{f}+\mathrm{f}_1$

Now, $0<\mathrm{f}<1$

$0<\mathrm{f}_1<1$

$0<\mathrm{f}+\mathrm{f}_1<2$

But as $198-[A]=f+f_1$, So, $f+f_1$ has to be an integer.

As, the only integer in $(0,2)$ is 1 , hence $f+f_1=1$

So, $[A]=197$

Q.10 Answer:

x

Solution:

We have $y=3x+6x^2+10x^3+\ldots\ldots$

$\Rightarrow 1+y=1+3x+6x^2+10x^3+\ldots.$

$\Rightarrow 1+y=(1-x)^{-3}\Rightarrow 1-x=(1+y)^{-1/3}$

$\Rightarrow x=1-(1+y)^{-1/3}$

$=\frac{1}{3}y-\frac{1\cdot4}{3^2\cdot2}y^2+\frac{1\cdot4\cdot7}{3^2\cdot3}y^3-\ldots..$

Q.11 Answer:

$-\frac{3}{8}x^2$

Solution:

$\because \mathrm{x}^3$ and higher powers of x may be neglected

$\because \frac{(1+x)^{\frac{3}{2}}-\left(1+\frac{x}{2}\right)^3}{(1-x)^{\frac{1}{2}}}$

$=(1-x)\frac{-1}{2}\left[\left(1+\frac{3}{2}x+\frac{\frac{3}{2}\cdot\frac{1}{2}}{2!}x^2\right)-\left(1+\frac{3x}{2}+\frac{3.2}{2!}\frac{x^2}{4}\right)\right.$

$=\left[1+\frac{x}{2}+\frac{\frac{1}{2}\cdot\frac{3}{2}}{2!}x^2\right]\left[\frac{-3}{8}x^2\right]=\frac{-3}{8}x^2$

(as x^3 and higher powers of x can be neglected)

Q.12 Answer:

9

Solution:

$27^{40}=3^{120}$

$3^{119}=(4-1)^{119}={}^{119}\mathrm{C}_0 4^{119}-{}^{119}\mathrm{C}_1 4^{118}$

$+{}^{119}\mathrm{C}_2 4^{117}-{}^{119}\mathrm{C}_3 4^{116}+\ldots.+(-1)$

$\therefore 3^{119}=4k-1$

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$\therefore 3^{120} = 12k - 3 = 12(k-1) + 9$

$\therefore$ The required remainder is 9.

Q.13 Answer:

7

Solution:

$(1073)^{71} - \mathrm{m} = (73 + 1000)^{71} - \mathrm{m}$

$= {}^{71}\mathrm{C}_0(73)^{71} + {}^{71}\mathrm{C}_1(73)^{70}(1000) + {}^{71}\mathrm{C}_2(73)^{69}$

$(1000)^2 + \ldots\ldots\ldots + {}^{71}\mathrm{C}_{71}(1000)^{71} - \mathrm{m}$

Above will be divisible by 10 if ${}^{71}\mathrm{C}_0(73)^{71}$ is divisible by 10

Now ${}^{71}\mathrm{C}_0(73)^{71} = (73)^{70} \cdot 73 = \left(73^2\right)^{35}.73$

The last digit of 73^2 is 9 , so the last digit of $\left(73^2\right)^{35}$ is 9 .

$\therefore$ Last digit of $\left(73^2\right)^{35}.73$ is 7

Hence, the minimum positive integral value of m is 7 , so that it is divisible by 10.

Q.14 Answer:

16

Solution:

$3^{2n+2} - 8n - 9, \forall n \in N$

Putting $n = 2$

$3^{2\times 2+2} - 8 \times 2 - 9 = 729 - 16 - 9 = 704$

It is divisible by 16

Q.15 Answer:

3

Solution:

$(1 + 0.0001)^{10000} = \left(1 + \frac{1}{n}\right)^n, n = 10000$

$= 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{2!}\frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!}\frac{1}{n^3} + \ldots\ldots$

$= 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \ldots\ldots$

$< 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots\ldots + \frac{1}{(9999)!}$

$= 1 + \frac{1}{1!} + \frac{1}{2!} + \ldots\ldots\infty = e < 3$

Q.16 Answer:

0

Solution:

We have

$7^9 + 9^7 = (8-1)^9 + (8+1)^7 = (1+8)^7 - (1-8)^9$

$= \left[1 + {}^7C_1 8 + {}^7C_2 8^2 + \ldots. + {}^7C_7 8^7\right] - \left[1 - {}^9C_1 8 + {}^9C_2 8^2 - \ldots.. - {}^9C_9 8^9\right]$

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$= {}^7C_1 8 + {}^9C_1 8 + \left[{}^7C_2 + {}^7C_3 \cdot 8 + \ldots. - {}^9C_2 + {}^9C_3 \cdot 8 - \ldots..\right] 8^2$

$= 8(7+9) + 64\mathrm{k} = 8.16 + 64\mathrm{k} = 64\mathrm{q}$

where $\mathrm{q} = \mathrm{k} + 2$

Thus, $7^9 + 9^7$ **is divisible by 64.**

Q.17 Answer:

$\frac{1}{15}$

Solution:

$\frac{2^{4n}}{15} = \frac{16^n}{15} = \frac{(1+15)^n}{15}$

$= \frac{1+{}^{\mathrm{n}}\mathrm{C}_1 15 + {}^{\mathrm{n}}\mathrm{C}_2 15^2 + \ldots. + {}^{\mathrm{n}}\mathrm{C}_{\mathrm{n}} 15^{\mathrm{n}}}{15}$

$= \frac{1+15\mathrm{k}}{15}$, **where** $\mathrm{k} \in \mathrm{N}, = \frac{1}{15} + \mathrm{k}$

$\therefore$ **Fractional part of** $\frac{2^{4n}}{15}$ **is** $\frac{1}{15}$

Q.18 Answer:

a natural number

Solution:

$\sqrt{5}\left[(\sqrt{5}+1)^{50} - (\sqrt{5}-1)^{50}\right]$

$= 2\sqrt{5}\left[{}^{50}\mathrm{C}_1(\sqrt{5})^{49} + {}^{50}\mathrm{C}_3(\sqrt{5})^{47} + \ldots..\right]$

$= 2\left[50\mathrm{C}_1(\sqrt{5})^{50} + {}^{50}\mathrm{C}_3(\sqrt{5})^{48} + \ldots.\right]$

= a natural number

Q.19 Answer:

1

Solution:

$x = (2+\sqrt{3})^n$

$= {}^nC_0 2^n + {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2}(\sqrt{3})^2 + \ldots$

Let $x_1 = (2-\sqrt{3})^n$

$= {}^nC_0 2^n - {}^nC_1 2^{n-1}\sqrt{3} + {}^nC_2 2^{n-2}(\sqrt{3})^2 + \ldots$

$x + x_1 = 2\left({}^nC_0 2^n + {}^nC_2 2^{n-2}(\sqrt{3})^2 + \ldots\right)$

$=$ **Even integer.**

Clearly, $x, \in (0,1) \forall \mathrm{n} \in \mathrm{N}$

$\Rightarrow [x] + \{x\} + x_1 =$ **Even integer**

$\Rightarrow \{x\} + x_1 =$ **Integer** $\{x\} \in (0,1), x_1 \in (0,1)$

$\Rightarrow \{x\} + x_1 \in (0,2)$

$\Rightarrow \{x\} + x_1 = 1 \Rightarrow x_1 = 1 - \{x\}$

$\Rightarrow x(1-\{x\}) = x \cdot x_1 = (2+\sqrt{3})^n (2-\sqrt{3})^n = 1$

Q.20 Answer:

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18

Solution:

We have, $7^{103} = 7(49)^{51} = 7(50-1)^{51}$

$= 7\left(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \ldots - 1\right)$

$= 7\left(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \ldots\right) - 7 + 18 - 18$

$= 7\left(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \ldots\right) - 25 + 18$

$= k + 18$ **(say)**

where k **is divisible by 25**

$\therefore$ **remainder is 18.**

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